

FRE6883 – Financial Computation

Recitation 1

C++ Fundamentals, Binomial Pricing, and Git

What are recitations for?

- **Complement the lectures** with additional explanations, examples, and practice.
- **Review the week's material**, focusing on the concepts and techniques introduced in class.
- **Ask questions** about anything that was unclear or deserves more practice.
- **Practice together** through exercises, coding examples, and debugging.

Come with your questions!

Recitations are here to support you throughout the course. Please bring any questions you have from the week's lectures, homework, or coding exercises.

You are also welcome to **send me your questions in advance** so that I can prepare examples or exercises around them.

What is your programming background?

Tell me which description fits.

1. Never programmed before
2. Python / R / Matlab only
3. Some C / C++
4. Took a full C++ course
5. Used C++ in a project or internship
6. Comfortable with pointers, references, and classes

Predict the output: copying a value

Assume `<iostream>` is included; this code is inside `main()`.

```
int x = 3;  
int y = x;  
y++;  
std::cout << x << " " << y;
```

Think, then vote

A 3 3

B 3 4

C 4 4

D Not sure

Does changing `y` change `x`?

Predict the output: copying a value

Assume `<iostream>` is included; this code is inside `main()`.

```
int x = 3;  
int y = x;  
y++;  
std::cout << x << " " << y;
```

Answer B: 3 4

y receives a copy of the value in x.

y++ increases y by one; x stays 3.

Predict the output: a short loop

Assume `<iostream>` is included; this code is inside `main()`.

```
for (int i = 0; i < 3; i++)  
{  
    std::cout << i << " ";  
}
```

Think, then vote

A 0 1 2

B 0 1 2 3

C 1 2 3

D Not sure

How many times does the body run?

Predict the output: a short loop

Assume `<iostream>` is included; this code is inside `main()`.

```
for (int i = 0; i < 3; i++)  
{  
    std::cout << i << " ";  
}
```

Answer A: 0 1 2

The body runs three times.

At `i = 3`, `i < 3` is false, so the loop stops.

A space follows each printed number.

What happens? A reference parameter

Optional diagnostic: new notation is welcome. Assume `<iostream>` is included.

```
void f(int& x)
{
    x++;
}
int main()
{
    int a = 3;
    f(a);
    std::cout << a;
    return 0;
}
```

Think, then vote

- A** Prints 3
- B** Prints 4
- C** Does not compile
- D** Not sure

Is x a copy of a here?

What happens? A reference parameter

Optional diagnostic: new notation is welcome. Assume `<iostream>` is included.

```
void f(int& x)
{
    x++;
}
int main()
{
    int a = 3;
    f(a);
    std::cout << a;
    return 0;
}
```

Answer B: prints 4

`int& x` makes `x` a reference to `a`: another name for the same variable.

Increasing `x` changes `a`.

Predict the output: integer division

Assume `<iostream>` is included; this code is inside `main()`.

```
int x = 5;  
double y = x / 2;  
std::cout << y;
```

Think, then vote

- A** 2
- B** 2.5
- C** Does not compile
- D** Not sure

Does declaring `y` as `double` change the division?

Predict the output: integer division

Assume `<iostream>` is included; this code is inside `main()`.

```
int x = 5;  
double y = x / 2;  
std::cout << y;
```

Answer A: prints 2

Both operands of `x / 2` are integers: the result is 2.

It is then stored as a double.

Use `x / 2.0` to obtain 2.5.

From a financial problem to an algorithm

A European call at expiry

Given the terminal stock price z and strike K , compute the payoff: $z - K$ if $z > K$, and zero otherwise.

Before writing C++:

1. What are the inputs and the output?
2. What steps should the program follow?
3. Which examples would convince you it works?

Next: describe the steps, then translate them into code.